

REG: A22

FIRST MID EXAMINATION, OCT/NOV- 2024**Course:** B.Tech III Year I SEM**Branch:** CSE/IT/AIML/DS/IOT**Subject:** COMPUTER NETWORKS**Time :** 2 Hours**Date :** 28.10.2024 (FN)**Max Marks:** 30**Part – A****(Compulsory Short answers)****6X 2 = 12 Marks**

S.N	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Discuss different types of topologies.	L2	CO1	2
2	From Unit-I	Compare various GUIDED transmission media in physical layer.	L2	CO1	2
3	From Unit-II	Draw frame format of HDLC and PPP	L3	CO2	2
4	From Unit-II	Explain Hamming code for error Detection and error correction.	L2	CO2	2
5	From Unit-III	Explain ALOHA	L2	CO3	2
6	From Unit-III	List types of bridges. And Write are the advantages of bridges	L1	CO3	2

Part – B**Answer any THREE out of the FOUR questions****3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Explain the functionality of each layer in the OSI reference model with neat sketch.	L2	CO1	6
8	From Unit-II	What are the different types of error detection methods? Explain the CRC error detection technique using generator polynomial $x^4 + x^2 + 1$ and data 1110001.	L1	CO2	6
9	From Unit-III	Explain CSMA/CD and CSMA/CA with flow charts.	L2	CO3	6
10 a)	From Unit-I	Explain network topologies	L2	CO1	2
b)	From Unit-II	Explain sliding window protocols	L1	CO2	2
c)	From Unit-III	Write short note on MAC addresses	L3	CO3	2

1245376

PAN
LW
WAG
JMAN

FIRST MID EXAMINATION, OCT/NOV- 2024

Branch: CSE/IT/DS/AIML/IOT

Time : 2 Hours

Max Marks: 30

(Compulsory Short answers)

S.N o	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	What is software engineering? Explain the layered technology?	L2	CO1	2
2	From Unit-I	Explain changing nature of the software's?	L2	CO1	2
3	From Unit-II	List out various types of Requirements.	L2	CO2	2
4	From Unit-II	List out the advantages and dis advantages of Waterfall model.	L2	CO2	2
5	From Unit-III	What is modeling? Write the Importance of modeling?	L1	CO3	2
6	From Unit-III	Explain about various types of Relationships with symbols?	L2	CO3	2

Answer any THREE out of the FOUR questions

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Explain in Detail about "The changing Nature of Software"?	L2	CO1	6
8	From Unit-II	Describe software Development Life Cycle with neat sketch?	L4	CO2	6
9	From Unit-III	Illustrate the conceptual model of UML?	L2	CO3	6
10 a)	From Unit-I	Identify various Umbrella activities in software process framework?	L3	CO1	2
b)	From Unit-II	Write the structure of Software Requirements Specification document.	L1	CO2	2
c)	From Unit-III	Explain about the Building blocks of UML?	L2	CO3	2

Course: B. Tech III Year I SEM**Branch:** CSE, IT, ECM**Subject:** Introduction to Data Science (9EC10)**Time :** 2 Hours**Date :** 30.10.2024 (FN)**Max Marks:** 30**Part – A****(Compulsory Short answers)****6X 2 = 12 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
1	Unit-I	Write the formula and a python code to calculate the standard deviation.	L1	CO1	2M
2	Unit-I	Differentiate nominal and ordinal data? Provide one example for each.	L4	CO1	2M
3	Unit-II	Write about any Four SET operations in Data Science.	L2	CO2	2M
4	Unit-II	State the Empirical Rule and its application in statistics.	L1	CO2	2M
5	Unit-III	Explain the concept of a sampling distribution and its significance in statistics.	L2	CO3	2M
6	Unit-III	How do a goodness-of-fit test and a test for independence differ in purpose and use?	L4	CO3	2M

Part – B**Answer any THREE out of the FOUR questions****3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	Unit-I	Discuss the importance of understanding the four levels of data when choosing statistical methods. Provide examples of two statistical techniques that are appropriate for ratio-level data.	L4	CO1	6M
8	Unit-II	Elaborate the 2 kinds of sampling techniques with examples.	L2	CO2	6M
9	Unit-III	For example, imagine a company wants to test the claim that their batteries last more than 20 hours. Using a simple random sample of 10 batteries yielded a mean of 21.5 hours, with a standard deviation of 6.2 hours. Test this claim using a significance level of 0.05.	L4	CO3	6M
10 a)	Unit-I	Identify the level of measurement for the following data Customer ID (1001, 1002, 1003, ...) Purchase Amount (₹) (1500, 2300, 4500, ...)	L1	CO1	2M
b)	Unit-II	What is Z-Score and explain its significance?	L2	CO2	2M
c)	Unit-III	List the steps for conducting a hypothesis test.	L1	CO3	2M

REG: A22



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

FIRST MID EXAMINATION, OCT/NOV- 2024

Course: B.Tech III Year I SEM

Subject: Entrepreneurship & Business Design

Date : 02.11.2024 (FN) (9ZC10)

Branch: CSE/IT/ECM

Time : 2 Hours

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	What is entrepreneurship?	L2	CO1	2
2	From Unit-I	Explain the difference between intrapreneur and entrepreneur	L2	CO1	2
3	From Unit-II	Describe about human centered design	L3	CO2	2
4	From Unit-II	Explain the different types of user interfaces	L1	CO2	2
5	From Unit-III	What are the warm and cool colors?	L1	CO3	2
6	From Unit-III	Explain briefly about customer journey	L3	CO3	2

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	Explain the multiple roles of an Entrepreneur	L1	CO1	6
8	From Unit-II	Compare the differences between UX and UI	L3	CO2	6
9	From Unit-III	Describe and explain the concept of color theory	L2	CO3	6
10 a)	From Unit-I	List out any two reasons feeding the entrepreneurial fire	L4	CO1	2
b)	From Unit-II	What are the types of prototyping strategies?	L1	CO2	2
c)	From Unit-III	Explain briefly on stage gate model	L2	CO3	2



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

FIRST MID EXAMINATION, OCT/NOV- 2024

Course: B. Tech. III Year I SEM

Branch: CSE / IT

Subject: Intellectual Property Rights (9LC04)

Time : 2 Hours

Date : 02/11/2024 (AN)

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Differentiate between Invention and Discovery.	L4	CO1	2
2	From Unit-I	What is Innovation? List out different techniques of Innovations.	L1,L4	CO1	2
3	From Unit-II	What is Patent Infringement? Name any two cases of Patent Infringement	L1,L4	CO1	2
4	From Unit-II	Describe the different types of Patent.	L1	CO1	2
5	From Unit-III	What is Copyright? What is the time-span for copyright protection.	L1	CO2	2
6	From Unit-III	What is Geographical Indications? Write some examples of Indian GIs.	L1,L4	CO2	2

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-I	What is Intellectual Property? Explain the different Intellectual Property Rights.	L1,L2	CO1	6
8	From Unit-II	Explain the Patentability Criteria.	L2	CO1	6
9	From Unit-III	Explain the Copyright registration application processing workflow.	L2	CO2	6
10 a)	From Unit-I	Write the different ways to transfer IP Rights to other?	L1	CO1	2
b)	From Unit-II	Write the different ways to search the patents.	L1	CO1	2
c)	From Unit-III	Explain the types of Copyright.	L2	CO2	2

to

FIRST MID EXAMINATION, OCT/NOV- 2024

Course: B.Tech III Year I SEM

Branch: CSE/ IT

Subject: Data Warehousing and Data Mining (9FC05)

Time: 2 Hours

Date : 04.11.2024(AN)

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-I	Define Data mining and Data Warehouse?	L1	CO1	2
2	From Unit-I	What is outlier analysis?	L1	CO1	2
3	From Unit-II	Why Preprocess the data?	L1	CO2	2
4	From Unit-II	Normalize the following group of data: 200, 300, 400, 600, 1000 by using min-max normalization setting min = 0 and max = 1	L4	CO2	2
5	From Unit-III	What are the differences between star schema and snowflake schema	L1	CO3	2
6	From Unit-III	Briefly explain about OLAP Operations	L1	CO3	2

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
11	From Unit-I	Describe the steps involved in KDD Process?	L2	CO1	6
12	From Unit-II	Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70. (a) What is the mean, median and mode of the data (b) Can you find the first quartile (Q1) and the third quartile (Q3) of the data? (c) Give the five-number summary of the data and Show a boxplot of the data.	L4	CO2	6
13	From Unit-III	Differentiate OLAP and OLTP	L4	CO3	6
14.a)	From Unit-I	What are the major issues in Data mining?	L1	CO1	2
b)	From Unit-II	How to handle the Missing data in Data Cleaning Process	L2	CO2	2
c)	From Unit-III	List out different types of OLAP servers	L1	CO3	2

14) (e) data mining is
inconsistent Data
noise data
redundancy

(b) using some universal keyword
entering manually

(c)

Ans 5) Star contains flat table + deterministic table
snowflake scheme non deterministic
of deterministic

□ snowflake reduce redundancy

Ans 6) Roll Back
Roll down
slide one
pivot

2 L311A0SL4

REG: A22



SCREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

SECOND MID EXAMINATION, DECEMBER 2024

Course: B.Tech III Year I SEM

Branch: CSE/IT/AIML/DS/IOT

Subject: Computer Networks -9EC05

Time : 2 Hours

Date : 28.12.2024 (FN)

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	Comparison of Virtual Circuit and Datagram Subnets	2	3	2
2	From Unit-IV	Discuss tunnel with benefits.	1	3	2
3	From Unit-V	Compare leaky bucket and token bucket	1	4	2
4	From Unit-V	Discuss keynote points of Sub netting and Super netting	1	4	2
5	From Unit-VI	Define Port. Examples Some Well-Known Ports	1	5	2
6	From Unit-VI	List out Socket API primitives.	1	5	2

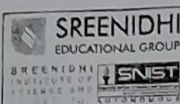
Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Discuss Distance Vector Routing with step-by-step diagrams	1	3	6
8	From Unit-V	Explain the IPV4 Header format	1	4	6
9	From Unit-VI	Discuss the TCP Connection State Machine	1	5	6
10 a)	From Unit-IV	Discuss the Link State Routing protocol	2	3	3
b)	From Unit-V	Discuss the IP address classes and their ranges	2	4	2
c)	From Unit-VI	Discuss keynote points of the Domain name system.	1	5	1

REG: A22



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnapet, Ghatkesar, Hyderabad – 501 301.

SECOND MID EXAMINATION, DEC- 2024

Course: B. Tech III Year I SEM

Branch: CSE/IT/DS/AI/ML/IOT

Subject: Software Engineering -9EC03

Time : 2 Hours

Date : 31.12.2024 (AN)

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	List out levels of Software Testing?	L1	CO4	2
2	From Unit-IV	Define Test Case. And write down the test cases for Validating ATM CARD.	L2	CO4	2
3	From Unit-V	Compute the code for the following metrics: if (k < 2) { if (k > 3) x = x*k; }	L6	CO5	2
4	From Unit-V	Find program level and effort	L4	CO5	2
5	From Unit-VI	Differentiate Flow Graph and Flowchart	L1	CO6	2
6	From Unit-VI	Abbreviate the term CRUD and EP?	L2	CO6	2
		What is Automation? Mention the advantages of Automation testing?			

Part – B

Answer any **THREE** out of the **FOUR** questions 3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Discuss in detail how the consequences of bugs are measured?	L3	CO4	6
8	From Unit-V	Explain in Details about Data flow testing with neat sketch	L2	CO5	6
9	From Unit-VI	Give the brief note for the following a) Alpha Testing b) Beta Testing c) Stress Testing	L1	CO6	6
10 a)	From Unit-IV	Specify the Characteristics of Test Engineers.	L2	CO4	2
b)	From Unit-V	Compare and contrast between Functional and Structural Testing.	L4	CO5	2
c)	From Unit-VI	Describe the Selenium Testing Tool?	L2	CO6	2

REG: A22



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

SECOND MID EXAMINATION, DEC- 2024

Course: B.Tech III Year I SEM

Branch: CSE/IT/ECM

Subject: Introduction to Data Science-9EC10

Time : 2 Hours

Date : 31.12.2024 (FN)

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.N	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	List out the differences between a Python list and a NumPy array?	L1	CO4	2
2	From Unit-IV	Explain the functionality of min () and max () functions in Pandas?	L2	CO4	2
3	From Unit-V	Outline the purpose of grid search in machine learning?	L1	CO5	2
4	From Unit-V	Discuss the concept of underfitting in machine learning?	L2	CO5	2
5	From Unit-VI	What is the difference between supervised and unsupervised learning?	L1	CO6	2
6	From Unit-VI	Define the concept of classification in machine learning?	L1	CO6	2

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Compare Pandas Series and Pandas DataFrame with an example?	L4	CO4	3
8	From Unit-V	Explain the bias-variance tradeoff in machine learning?	L2	CO5	3
9	From Unit-VI	What is a machine learning decision tree, and explain its main components?	L2	CO6	3
10 a)	From Unit-IV	Calculate the sum of all values in a specific column across all rows in a DataFrame using the sum () function as an example?	L3	CO4	1
b)	From Unit-V	Evaluate the role of K-fold cross-validation in the performance of machine learning models?	L5	CO5	1
c)	From Unit-VI	Differentiate between machine learning and traditional programming?	L2	CO6	1

Course: B.Tech III Year I SEM**Branch:** CSE/ ECM/ IT**Subject:** Entrepreneurship Business Design-- 9ZC10**Time :** 2 Hours**Date :** 03 .01 .2025 (FN)**Max Marks:** 30**Part – A**

(Compulsory Short answers)

6X 2 = 12 Marks

S.N o	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	How services are delivered?	L2	CO4	2M
2	From Unit-IV	Explain the concept of branding for services	L2	CO4	2M
3	From Unit-V	Discuss any two pricing strategies	L1	CO5	2M
4	From Unit-V	What is a solution Demo?	L2	CO5	2M
5	From Unit-VI	What is Break Even Analysis?	L1	CO6	2M
6	From Unit-VI	Explain the concept of Risk Propensity	L2	CO6	2M

Part – BAnswer any **THREE** out of the FOUR questions**3X 6 = 18 Marks**

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Discuss the concept of service flow and explain core services and enhancing services	L3	CO4	6M
8	From Unit-V	What is a Business Model? Explain Types of Business Models	L2	CO5	6M
9	From Unit-VI	Discuss various sources of funding for Entrepreneurs	L3	CO6	6M
10	From Unit-IV	List out any two types of new services	L1	CO4	2M
a)	From Unit-V	What is a Minimum viable Product (MVP)	L2	CO5	2M
b)	From Unit-VI	Explain the concept of Locus of Control	L2	CO6	2M

SL4

REG: A22



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

SECOND MID EXAMINATION, JANUARY 2024

Course: B.Tech III Year I SEM

Subject: IPR-9IC04

Date : 03.01.2024 (AN)

Branch: CSE/IT

Time : 2 Hours

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.N	Coverage		BCLL	CO(s)	Marks
0					
1	From Unit-IV	Define certification mark and Service mark?	L1	CO4	2M
2	From Unit-IV	Define Trade Secret? Write the Examples of Trade secret.	L1	CO4	2M
3	From Unit-V	What is the Eligibility for design protection?	L1	CO5	2M
4	From Unit-V	Define Infringement? Write the Example of Industrial Design Infringement.	L1	CO5	2M
5	From Unit-VI	Summarize the role of Patent Cooperation Treaty?	L2	CO6	2M
6	From Unit-VI	Explain about Berne Convention	L2	CO6	2M

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks
7	From Unit-IV	Distinguish between trademark and patent?	L2	CO4	6
8	From Unit-V	Explain the need for protection of integrated circuits and layout design?	L6	CO5	6
9	From Unit-VI	Discuss the functions and agreements of World Trade organization?	L2	CO6	6
10	From Unit-IV	Explain the advantages of Trademark registration?	L2	CO4	2
a)	From Unit-V	Explain the importance of filing for Industrial Design?	L2	CO5	2
b)	From Unit-VI	Describe the objectives of GATT?	L1	CO6	2
c)					

2231A0SL9

REG: A22



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad – 501 301.

SECOND MID EXAMINATION, JAN 2025

Course: B.Tech III Year I SEM

Branch: CSE/ IT/AI/ML

Subject: Data Warehousing and Data Mining -9FC05

Time : 2 Hours

Date : 04.01.2025 - AN

Max Marks: 30

Part – A

(Compulsory Short answers)

6X 2 = 12 Marks

S.No	Coverage	PLAUS	PLB _R	PLC	BCLL	CO(s)	Marks
1	Unit-IV	What is support and confidence? Give the formulas			L1	CO4	2M
2	Unit-IV	Explain why the FP-Growth algorithm is preferred over the Apriori algorithm			L2	CO4	2M
3	Unit-V	List attributes selection measures. write the formula for each of them			L1	CO5	2M
4	Unit-V	Compare classification and prediction. provide a list of methods commonly used for both classification and prediction			L1	CO5	2M
5	Unit-VI	What is ordinal and nominal data? Give examples			L1	CO6	2M
6	Unit-VI	What is the criteria of good cluster			L1	CO6	2M

Part – B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

S.No	Coverage		BCLL	CO(s)	Marks										
7	From Unit-IV	Given the database, apply the Apriori algorithm to identify strong association rules, using a minimum support of 50% and a minimum confidence of 80%. <table><tr><th>TID</th><th>Items</th></tr><tr><td>100</td><td>1 3 4</td></tr><tr><td>200</td><td>2 3 5</td></tr><tr><td>300</td><td>1 2 3 5</td></tr><tr><td>400</td><td>2 5</td></tr></table>	TID	Items	100	1 3 4	200	2 3 5	300	1 2 3 5	400	2 5	L3	CO4	6M
TID	Items														
100	1 3 4														
200	2 3 5														
300	1 2 3 5														
400	2 5														
8	From Unit-V	What is classification? Explain the classification as a two step process with example.	L2	CO5	6M										
9	From Unit-VI	What is density-based clustering? Explain the DBSCAN algorithm	L3	CO6	6M										
10 a)	From Unit-IV	Discuss multi level association rules	L2	CO4	2M										
b)	From Unit-V	Briefly explain KNN algorithm?	L2	CO5	2M										
c)	From Unit-VI	Write K-means Algorithm?	L2	CO6	2M										

Sup(AUD)
son(A)



SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY
Autonomous Institution
Yamnampet, Ghatkesar, Hyderabad - 501 301.

SECOND MID EXAMINATION, DECEMBER 2024

Course: B.Tech III Year I SEM

Branch: CSE/IT/ECM/CS

Subject: Design and Analysis of Algorithms -9EC04

Time : 2 Hours

Date : 28.12.2024 (AN)

Max Marks: 30

Part - A

(Compulsory Short answers)

6X 2 = 12 Marks

S.N	Coverage		BCLL	CO(s)	Marks
1	From Unit-IV	State the principle of optimality in Dynamic Programming?	L1	CO4	2M
2	From Unit-IV	List out the applications of Dynamic Programming?	L1	CO4	2M
3	From Unit-V	Define Live node, E-node and Dead node. ✓	L1	CO5	2M
4	From Unit-V	What is Hamiltonian Cycle with an example? ✓	L1	CO5	2M
5	From Unit-VI	Give an overview of LCBB and FIFOBB. ✓	L2	CO6	2M
6	From Unit-VI	Write the differences between NP Hard and NP Complete problems. ✓	L1	CO6	2M

Part - B

Answer any THREE out of the FOUR questions

3X 6 = 18 Marks

3X 6 = 18 Marks

Answer any <u>THREE</u> out of the <u>FOUR</u> questions			BCLL	CO(s)	Marks												
7	From Unit-IV	Solve the given data using Matrix Chain multiplication in dynamic programming. $A(2,3)*B(3,4)*C(4,2)$	L3	CO4	6M												
8	From Unit-V	Explain about Backtracking with the help of n-queens problem	L2	CO5	6M												
9	From Unit-VI	Solve the following Knapsack using the LC branch and bound technique with knapsack capacity, $W=6$. <table border="1" data-bbox="614 1308 991 1458"> <tr> <td>Items</td> <td>1</td> <td>2</td> <td>3</td> </tr> <tr> <td>W_i</td> <td>2</td> <td>3</td> <td>2</td> </tr> <tr> <td>P_i</td> <td>8</td> <td>6</td> <td>4</td> </tr> </table>	Items	1	2	3	W_i	2	3	2	P_i	8	6	4	L3	CO6	6M
Items	1	2	3														
W_i	2	3	2														
P_i	8	6	4														
10 a)	From Unit-IV	What is a knapsack problem? How dynamic programming is useful for solving it?	L1	CO4	2												
b)	From Unit-V	Define m-coloring problem with an example.	L2	CO5	2												
c)	From Unit-VI	Differentiate NP-Hard and NP-Complete problems.	L4	CO6	2												

Code No:9EC05

Date:16 -January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)
COMPUTER NETWORKS (CSE,IT,AI ML,DS & IOT)

Time: 3 Hours

Max.Marks:60

Note: a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 1. What is a computer network? Write uses of networks. | L2 | CO1 | [2M] |
| 2. Write design issues of the data link layer. | L1 | CO2 | [2M] |
| 3. What are the uses of bridges in networks? | L2 | CO3 | [2M] |
| 4. Distinguish between broadcasting and multicasting. | L2 | CO4 | [2M] |
| 5. List principles of congestion. | L2 | CO4 | [2M] |
| 6. List protocols of the application layer. | L1 | CO5 | [2M] |

Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 7. Explain OSI Reference Model in detail.
OR | L2 | CO1 | [8M] |
| 8. Describe fiber optics. | L2 | CO1 | [8M] |
| 9. Explain Go Back-N Sliding Window protocol with neat diagram
OR | L2 | CO2 | [8M] |
| 10. Illustrate the calculation of CRC code for the frame 110101 using the generator $G(x) = x^3 + x + 1$. | L3 | CO2 | [8M] |
| 11. Describe CSMA protocols with neat sketch.
OR | L2 | CO3 | [8M] |
| 12. Summarize various types of bridges. | L4 | CO3 | [8M] |
| 13. Illustrate shortest path routing with an example.
OR | L3 | CO4 | [8M] |
| 14. Explain the link state routing algorithm with an example. | L3 | CO4 | [8M] |
| 15. Draw IPV4 header format and explain each field.
OR | L4 | CO5 | [8M] |
| 16. Describe congestion prevention policies. | L2 | CO5 | [8M] |
| 17. Illustrate TCP connection establishment with suitable figure.
OR | L4 | CO5 | [8M] |
| 18. Explain domain name system in detail. | L2 | CO5 | [8M] |

-- 00 -- 00 --

Code No:9EC03

Date: 23-January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)
SOFTWARE ENGINEERING (CSE,IT,AI ML,DS & IOT)

Time: 3 Hours

Max.Marks:60

Note: a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 1 List all the umbrella activities in Process framework? | L2 | CO1 | [2M] |
| 2 What is the need for System Requirements? | L2 | CO1 | [2M] |
| 3 List the principles of a Software Design? | L2 | CO2 | [2M] |
| 4 Define validation in Software Testing? | L2 | CO3 | [2M] |
| 5 Define Transaction Flow graph? | L2 | CO3 | [2M] |
| 6 What is the use of White Box Testing? | L2 | CO3 | [2M] |

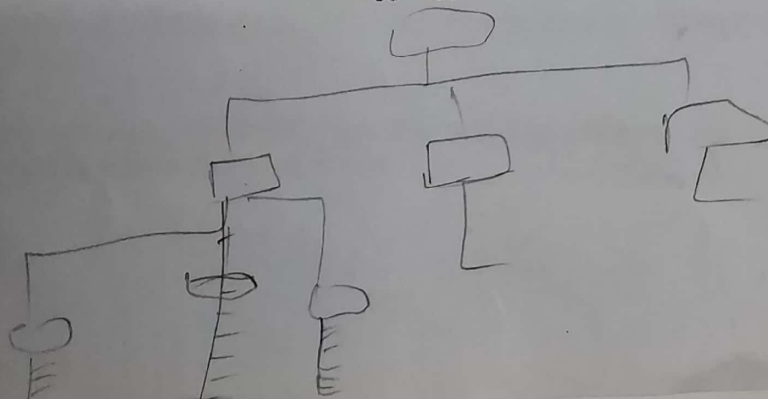
Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 7 Discuss various types of Software myths and true aspects of these myths?
OR | L2 | CO1 | [8M] |
| 8 Elaborate the changing nature of Software? | L2 | CO1 | [8M] |
| 9 Discuss about Functional and Non-Functional Requirements?
OR | L2 | CO2 | [8M] |
| 10 Summarize Agile development in Software Engineering?
OR | L4 | CO2 | [8M] |
| 11 Demonstrate the uses of Use cases in Requirements elicitation and analysis with an example?
OR | L4 | CO2 | [8M] |
| 12 Define Software Architecture? Explain Conceptual model of the UML ? | L4 | CO2 | [8M] |
| 13 Describe various Testing principles? Discuss about software Testing Life Cycle?
OR | L2 | CO3 | [8M] |
| 14 What are Test coverage criteria? Discuss Testing issues? | L2 | CO3 | [8M] |
| 15 Define predicates? Explain multi way branches and inputs used in Path Testing?
OR | L3 | CO3 | [8M] |
| 16 Define software quality? What are the metrics for analysis model? | L2 | CO3 | [8M] |
| 17 Differentiate between boundary values analysis and Regression Testing?
OR | L4 | CO3 | [8M] |
| 18 Discuss about Selenium Testing Tool in brief? | L3 | CO3 | [8M] |

-- 00 -- 00 --



Code No:9EC10

Date:21-January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)

INTRODUCTION TO DATA SCIENCE (PE-I) (CSE,IT & ECM)

Time: 3 Hours

Max.Marks:60

- Note:** a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

	BCLL	CO(s)	Marks
1 Differentiate Structured and Unstructured Data	L2	CO1	[2M]
2 What is the coefficient of variation, and when is it used?	L1	CO2	[2M]
3 What is a Type I error in hypothesis testing?	L1	CO2	[2M]
4 Define a Pandas Data Frame and its primary use in data analysis.	L1	CO3	[2M]
5 Define cross-validation and its purpose in model evaluation.	L1	CO4	[2M]
6 Define supervised learning and provide one example	L1	CO5	[2M]

Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

	BCLL	CO(s)	Marks
7. Discuss the four levels of data and their implications in data science	L2	CO1	[8M]
OR			
8 Explain the importance of understanding data types before applying statistical and machine learning models.	L3	CO1	[8M]
9. Illustrate the process of matrix multiplication and its role in recommendation systems.	L3	CO2	[8M]
OR			
10 Discuss the importance of normalization using z-scores and its impact on data comparability	L2	CO2	[8M]
11 Explain the concept of sampling distributions and their relevance in statistical analysis.	L3	CO2	[8M]
OR			
12 Explain the chi-square test for independence and its application in assessing relationships between variables.	L2	CO2	[8M]
13 Explain the concept of aggregation in NumPy and Pandas, including its importance in summarizing data.	L3	CO3	[8M]
OR			
14 Illustrate the process of merging and joining datasets in Pandas, highlighting their practical applications.	L3	CO3	[8M]
15 Discuss the challenges associated with high bias and high variance and discuss the 2 extreme cases of bias/variance tradeoff.	L2	CO4	[8M]

OR

- 16 Describe the process of K- fold cross validation and grid searching. L2 CO4 [8M]
- 17 Discuss the different types of machine learning with examples L2 CO5 [8M]
- OR
- 18 Determine how supervised learning models are evaluated using metrics like accuracy, precision, and recall. L3 CO5 [8M]

-- 00 -- 00 --





H.T No

Sreenidhi Institute of Science and Technology
(An Autonomous Institution)

Regulations:
A22

Code No: 9ZC10

Date: 25-January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)
ENTREPRENEURSHIP AND BUSINESS DESIGN(CSE, IT, ECM and DS)

Time: 3 Hours

Max.Marks:60

Note: a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 1 Who is an Intrapreneur? | L1 | CO1 | [2M] |
| 2 Define responsive design and mention its benefits. | L1 | CO2 | [2M] |
| 3 What is journey mapping in the UX process? | L1 | CO3 | [2M] |
| 4 What is the Service Flower model in service design? | L1 | CO4 | [2M] |
| 5 What is the purpose of the Business Model Canvas? | L1 | CO5 | [2M] |
| 6 What is the difference between risk propensity and risk avoidance? | L1 | CO6 | [2M] |

Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

- | | BCLL | CO(s) | Marks |
|---|------|-------|-------|
| 7 Explain the entrepreneurial process and describe why it is essential for economic development. | L1 | CO1 | [8M] |
| OR | | | |
| 8 Identify an entrepreneur in your local area. Analyze which of the multiple roles (intrapreneur, inventor, coordinator, manager, controller) they embody the most. | L4 | CO1 | [8M] |
| 9 Explain the concept of human-centered design and its benefits in UI/UX development. | L1 | CO2 | [8M] |
| OR | | | |
| 10 Evaluate the effectiveness of a responsive design for an e-commerce website. How does it enhance the overall user experience? | L4 | CO2 | [8M] |
| 11 What are the key steps involved in designing a business strategy, including defining a problem, mapping user journeys, and pitching ideas? | L1 | CO3 | [8M] |
| OR | | | |
| 12 Using a Stage-Gate Model, outline how you would evaluate and kill ideas that may not be feasible for a start-up. | L4 | CO3 | [8M] |
| 13 Define the concept of "services as solutions" and list the key elements of a service flower. | L1 | CO4 | [8M] |
| OR | | | |
| 14 Explain the rapid branding and marketing strategies. | L4 | CO4 | [8M] |
| 15 Define the term "business model" and list its key components. | L1 | CO5 | [8M] |
| OR | | | |
| 16 Identify and analyze the risks and assumptions in a subscription-based software business model. How might these impact the business? | L4 | CO5 | [8M] |



- 17 Define **bootstrapping**, **angel investors**, and **venture capitalists** in the context of entrepreneurial funding. L1 CO6 [8M]

OR

- 18 Analyze the relationship between **customer acquisition costs** and **return on equity (ROE)** for a startup. How do high customer acquisition costs affect profitability? L4 CO6 [8M]

-- 00 -- 00 --

Code No: 9FC05

Date: 30-January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)

DATA WAREHOUSING AND DATA MINING (CSE, IT and AIML)

Time: 3 Hours

Max.Marks:60

- Note:** a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 1. What is the need for data integration? | L2 | CO1 | [2M] |
| 2. Give concept hierarchy for address location. | L1 | CO2 | [2M] |
| 3. List the Data warehouse Characteristics. | L2 | CO3 | [2M] |
| 4. Define multilevel association rules in data mining. | L1 | CO4 | [2M] |
| 5. What is classification? | L2 | CO5 | [2M] |
| 6. What are the types of Data in cluster Analysis | L1 | CO5 | [2M] |

Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

- | | BCLL | CO(s) | Marks |
|--|------|-------|-------|
| 7. What is KDD? Explain about data mining as a step in the process of knowledge discovery. | L4 | CO1 | [8M] |
| OR | | | |
| 8. Discuss in detail about Data Mining Functionalities | L2 | CO1 | [8M] |
| 9. Describe the process and importance of descriptive data summarization in data preprocessing. | L2 | CO2 | [8M] |
| OR | | | |
| 10. Explain the steps involved in data integration and transformation with suitable examples. | L4 | CO2 | [8M] |
| 11. What is the significance of OLAP in data warehouse? Describe OLAP operations with necessary diagram/example. | L4 | CO3 | [8M] |
| OR | | | |
| 12. With a neat sketch, Explain architecture of data ware housing. | L3 | CO3 | [8M] |
| 13. Discuss the methods for mining multilevel association rules from transaction databases. | L2 | CO4 | [8M] |
| OR | | | |
| 14. Compare and contrast multilevel and multidimensional association rule mining techniques. | L2 | CO4 | [8M] |
| 15. a) What are the characteristics of k-nearest neighbor algorithm? | L4 | CO5 | [8M] |
| b) How to evaluate the classifier accuracy? | L5 | | |
| OR | | | |
| 16. Illustrate how Support Vector Machines (SVMs) are used for classification | L3 | CO5 | [8M] |
| 17. List the density based clustering methods. discuss DBSCAN clustering method in detail. | L3 | CO5 | [8M] |
| OR | | | |
| 18. Compare and contrast text mining and web mining, highlighting their similarities and differences. | L5 | CO6 | [8M] |

-- 00 -- 00 --

Handwritten diagram showing a coordinate system with axes labeled 'SUM' and 'COO'. There are several small circles plotted in the first quadrant, and a line of best fit is drawn through them.

Code No:9EC04

Date: 18-January-2025 (AN)

B.Tech III-Year I- Semester External Examination, January-2025 (Regular)

DESIGN AND ANALYSIS OF ALGORITHMS (CSE, IT, ECM & CS)

Time: 3 Hours

Max.Marks:60

Note: a) No additional answer sheets will be provided.
b) All sub-parts of a question must be answered at one place only, otherwise it will not be valued.
c) Missing data can be assumed suitably.

Bloom's Cognitive Levels of Learning (BCLL)

Remember	L1	Apply	L3	Evaluate	L5
Understand	L2	Analyze	L4	Create	L6

Part - A

Max.Marks: 6x2=12

ANSWER ALL QUESTIONS, EACH QUESTION CARRIES 2 MARKS.

- | <ol style="list-style-type: none"> 1 What is Space complexity? 2 Give the time complexities of Merge sort and Binary search. 3 What do you mean by Greedy method? 4 List the features of dynamic programming. 5 Define State space tree. 6 Define P and NP. | <table border="0"> <tr> <th>BCLL</th><th>CO(s)</th><th>Marks</th></tr> <tr> <td>L2</td><td>CO1</td><td>[2M]</td></tr> <tr> <td>L3</td><td>CO2</td><td>[2M]</td></tr> <tr> <td>L2</td><td>CO3</td><td>[2M]</td></tr> <tr> <td>L1</td><td>CO3</td><td>[2M]</td></tr> <tr> <td>L2</td><td>CO4</td><td>[2M]</td></tr> <tr> <td>L2</td><td>CO4</td><td>[2M]</td></tr> </table> | BCLL | CO(s) | Marks | L2 | CO1 | [2M] | L3 | CO2 | [2M] | L2 | CO3 | [2M] | L1 | CO3 | [2M] | L2 | CO4 | [2M] | L2 | CO4 | [2M] |
|---|---|-------|-------|-------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|
| BCLL | CO(s) | Marks | | | | | | | | | | | | | | | | | | | | |
| L2 | CO1 | [2M] | | | | | | | | | | | | | | | | | | | | |
| L3 | CO2 | [2M] | | | | | | | | | | | | | | | | | | | | |
| L2 | CO3 | [2M] | | | | | | | | | | | | | | | | | | | | |
| L1 | CO3 | [2M] | | | | | | | | | | | | | | | | | | | | |
| L2 | CO4 | [2M] | | | | | | | | | | | | | | | | | | | | |
| L2 | CO4 | [2M] | | | | | | | | | | | | | | | | | | | | |

Part - B

Max.Marks: 6x8=48

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 8 MARKS.

- | <ol style="list-style-type: none"> 7 Discuss various the asymptotic notations used for best case average case and worst case analysis of algorithms. <p align="center">OR</p> <ol style="list-style-type: none"> 8 Solve the recurrence relation and compute its complexity.
 $T(1)=1$
 $T(n) = 2T(n-1) + n, n > 1$ <p align="center">OR</p> <ol style="list-style-type: none"> 9 Describe the algorithm for <i>Quick sort</i> using <i>divide and conquer</i> with an example. <p align="center">OR</p> <ol style="list-style-type: none"> 10 Illustrate the <i>Strassen's matrix multiplication</i> with an example. <p align="center">OR</p> <ol style="list-style-type: none"> 11 Find the optimal solution for the Knapsack instance $n = 7, m = 15$; $(p_1, p_2, p_3, p_4, p_5, p_6, p_7) = (10, 5, 15, 7, 6, 18, 3)$ and $(w_1, w_2, w_3, w_4, w_5, w_6, w_7) = (2, 3, 5, 7, 1, 4, 1)$ by using Greedy approach. <p align="center">OR</p> <ol style="list-style-type: none"> 12 Describe job sequencing with deadlines problem and Find the solution generated by job sequencing deadlines when $n=5$, $(p_1, p_2, p_3, p_4, p_5) = (10, 3, 33, 11, 40)$ and $(d_1, d_2, d_3, d_4, d_5) = (3, 1, 1, 2, 2)$. <p align="center">OR</p> <ol style="list-style-type: none"> 13 Illustrate the travelling sales person problem and discuss how to solve it by using dynamic programming. <p align="center">OR</p> <ol style="list-style-type: none"> 14 Determine the solution for 0/1 knapsack problem using dynamic programming $N=3, m=6$ profits $(p_1, p_2, p_3) = (1, 2, 5)$ weights $(w_1, w_2, w_3) = (2, 3, 4)$. <p align="center">OR</p> <ol style="list-style-type: none"> 15 Explain 4-Queens problem using backtracking and Draw the state space tree for 4-Queens problem. | <table border="0"> <tr> <th>BCLL</th><th>CO(s)</th><th>Marks</th></tr> <tr> <td>L3</td><td>CO1</td><td>[8M]</td></tr> <tr> <td>L4</td><td>CO1</td><td>[8M]</td></tr> <tr> <td>L2</td><td>CO2</td><td>[8M]</td></tr> <tr> <td>L3</td><td>CO2</td><td>[8M]</td></tr> <tr> <td>L4</td><td>CO3</td><td>[8M]</td></tr> <tr> <td>L2</td><td>CO3</td><td>[8M]</td></tr> <tr> <td>L2</td><td>CO3</td><td>[8M]</td></tr> <tr> <td>L3</td><td>CO4</td><td>[8M]</td></tr> <tr> <td>L3</td><td>CO4</td><td>[8M]</td></tr> </table> | BCLL | CO(s) | Marks | L3 | CO1 | [8M] | L4 | CO1 | [8M] | L2 | CO2 | [8M] | L3 | CO2 | [8M] | L4 | CO3 | [8M] | L2 | CO3 | [8M] | L2 | CO3 | [8M] | L3 | CO4 | [8M] | L3 | CO4 | [8M] |
|---|--|-------|-------|-------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|----|-----|------|
| BCLL | CO(s) | Marks | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L3 | CO1 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L4 | CO1 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L2 | CO2 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L3 | CO2 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L4 | CO3 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L2 | CO3 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L2 | CO3 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L3 | CO4 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| L3 | CO4 | [8M] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

$Q_1(1, 3)$
 $Q_2(2, 4)$

$P_n(1, 2, 3, 4)$

- 16 Illustrate the usage of backtracking to find all Hamiltonian cycles. L2 CO4 [8M]
- 17 Construct the portion of state space tree generated for the following 0/1 Knapsack problem instance using FIFO Branch and Bound. L3 CO4 [8M]
(p1, p2, p3, p4, p5) = (10, 15, 6, 8, 4), (w1, w2, w3, w4, w5) = (4, 6, 3, 4, 2), M = 12, n = 5.

OR

- 18 Illustrate the Non deterministic algorithms in detail. L2 CO4 [8M]

-- 00 -- 00 --